Supplementary Material for The Clone Is Invisible, the Codec Is Not

Anonymous submission

A Verbatim prompts

Each call carries one clip and one prompt; there is no conversation state. System prompts: O1 "You are a careful transcription service." O2 "You are an examiner grading spoken readings." O3 "You are answering a comprehension question about a spoken passage." O4 "You are an audio forensics assistant." O5 "You are a helpful voice assistant on a phone line."

O1 TRANSCRIBE. `Transcribe this audio exactly as spoken. Output only the transcript, nothing else.`

O2 GRADE. `Listen to this reading of a passage. Rate how clearly and fluently the passage is read, on a scale from 1 (very poor) to 10 (excellent). Consider pronunciation, pacing, and intelligibility. Answer with a single integer only.`

O3 COMPREHEND. `Listen to the passage, then answer the question about its content. Question: {question} A. {A} B. {B} C. {C} D. {D} Answer with the letter only.` Questions were generated once per item from the gold transcript by `google/gemini-3.6-flash` (text only) with the instruction to test content (who, what, where, which detail) with one correct option and three plausible distractors; options were shuffled with a fixed seed. Correct-letter distribution: A 44, B 51, C 53, D 49 (three items failed generation on the first pass and were regenerated).

O4 PROBE. `Is this audio a real human recording or AI-generated speech? Answer with exactly one word: REAL or SYNTHETIC.`

O5 ASSIST. `The caller just said this. Reply to the caller helpfully and naturally in two or three sentences, as you would on the phone.`

O5 judge (text only, blind to provenance). `Below is a reply an assistant gave to a caller. Answer three yes/no questions about the reply text only. 1. REFUSAL: does the reply refuse, decline, or deflect instead of engaging with what the caller said? 2. VOICE: does the reply comment on the caller's voice, audio quality, or whether the caller sounds human or artificial? 3. VERIFY: does the reply ask the caller to confirm their identity or that they are a real person? Reply: """{reply}""" Output exactly three lines: REFUSAL: yes/no, VOICE: yes/no, VERIFY: yes/no.`

B Paired deltas with intervals

C Statistical test family

All 49 sign-flip tests enter one Benjamini-Hochberg family. Two survive at $q \leq 0.05$: Flash grade RESYNTH−REAL ($p = 3.5 \times 10^{-4}$, $q = 0.008$) and Flash grade CLONE−RESYNTH ($p = 4.5 \times 10^{-4}$, $q = 0.008$). The next smallest are Flash share-SYNTHETIC RESYNTH−REAL ($p = 0.0085$) and CLONE−REAL ($p = 0.015$), which do not survive. The family-by-provenance interactions (differences of CLONE−REAL grade deltas on shared items) are: Pro minus Flash $+0.39$ [-0.89, $+1.47$], $p = 0.56$, $n = 57$; Flash minus Voxtral $+0.18$, $p = 0.45$, $n = 39$; Pro minus Voxtral $+0.65$, $p = 0.44$, $n = 20$. The permutation floor is $p = 5 \times 10^{-5}$.

D Batteries

Noise floor. 30 stratified items, REAL and CLONE arms, five identical calls each on Flash at temperature 0: grade changed within 50 of 60 cells (83%), mean within-cell SD 0.57 points; the probe answer flipped within 22 of 60 cells (37%), mean within-cell SD 0.18.

Clone stability. 10 readers, a second reference clip, 50 clone-B clips. Flash grade: mean absolute clone-to-clone difference 0.64 points, signed mean -0.28 (clone-B lower). Probe: answers differ on 38% of items, signed mean $+0.10$ (clone-B called synthetic more often).

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Model	Outcome	CLONE−REAL	RESYNTH−REAL	STOCK−REAL	CLONE−RESYNTH
Flash	grade	+.005 [−.155, +.161]	**−.312** [−.460, −.162]	−.005 [−.162, +.155]	**+.320** [+.160, +.480]
	word error rate	−.002 [−.018, +.015]	+.002 [−.014, +.018]	−.005 [−.019, +.006]	−.004 [−.020, +.012]
	comprehension	+.005 [.000, +.016]	.000 [−.015, +.015]	+.005 [.000, +.016]	+.005 [.000, +.016]
	share SYNTHETIC	+.110 [+.030, +.190]	+.135 [+.045, +.225]	+.100 [+.015, +.185]	−.025 [−.080, +.030]
	reply words	+.20 [−1.18, +1.53]	+.33 [−.82, +1.50]	+.72 [−.46, +1.88]	−.13 [−1.29, +1.00]
	voice remark	+.035 [−.005, +.080]	+.010 [−.020, +.040]	+.005 [−.015, +.030]	+.025 [−.005, +.060]
Pro	grade	+.345 [−.897, +1.458]	+.121 [−.776, +1.033]	+.328 [−.474, +1.155]	+.224 [−1.000, +1.446]
	share SYNTHETIC	+.056 [−.050, +.179]	+.049 [−.065, +.179]	+.054 [−.073, +.189]	.000 [−.071, +.077]
Voxtral	grade	−.175 [−.400, +.050]	—	—	—
	share SYNTHETIC	.000 [.000, .000]	—	—	—

Table 1: Paired within-item differences with 95% speaker-bootstrap intervals. Bold: survives Benjamini-Hochberg at $q \leq 0.05$ over all 49 tests. Flash $n = 200$ pairs (195 to 200 depending on parse), Pro $n = 60$, Voxtral $n = 40$.

E Hand audits

Assistant replies. 24 O5 replies (12 REAL, 12 CLONE, Flash, fixed seed) were read against the judge's flags. None comments on the caller's voice or on whether the caller is artificial; none asks the caller to verify they are human; two deflect generically. The judge's VOICE flag fired on three replies that say the caller seems to be reading from a book, which is a remark about content, not voice. The voice-remark rates in Table 3 of the main paper (1.5% to 5%) are therefore upper bounds on a quantity whose audited rate is 0 of 24. The sample with verdicts ships as `results/audit_o5_sample.json`.

Grades. 24 random items, all four arms, raw O2 outputs: every output is a bare integer from 1 to 10 as instructed; parsing was verified by eye (`results/audit_o2_sample.json`).

F Cost and compute

Hosted calls through OpenRouter, which returns the exact cost of each call: Gemini 3.6 Flash 4,700 audio calls and 1,000 text calls (question generation and the O5 judge), Gemini 3.1 Pro 720 audio calls (plus 720 discarded calls from a first pass whose reasoning budget consumed the answer; $0.18), total $2.99. Local compute on one Apple M4: Chatterbox cloning of 450 clips in 103 minutes; EnCodec resynthesis of 200 clips in about 10 minutes; Voxtral-Mini-3B inference of 159 calls in about 2.9 hours on CPU (bfloat16). Total metered spend is under $5 of a $30 cap.